

DrugBank Database: Data Pipeline and Exploratory Analysis (Weeks 4 -6)

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ITAI 2277 - SPRING 2026

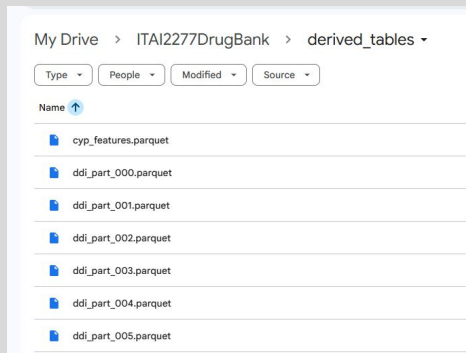
DrugBank Dataset : Data Acquisition and Collection

Selected because it aligns with project goal:
drug combination safety analysis

Source provided as a **1.76 GB XML file**,
processed using structure parsing in
Google Colab

Stream-parse the XML and extract **ONLY**
what we need. This creates 4 compact
tables that align with CompoundIQ needs

Stored outputs as
Parquet files for
faster access and
reduced storage.



Raw biomedical XML
successfully transformed into
structured, reusable datasets.

TABLE	ROWS	SIZE	PURPOSE
drugs.parquet	1,014,348	5.31 MB	Core drug metadata (from nested XML extraction)
drug_interactions.parquet	2,910,010	55.15 MB	Drug-drug interaction relationships
targets.parquet	24,231	0.56 MB	Drug-target associations
enzymes.parquet	6,062	0.08 MB	Metabolic enzyme (CYP) data

DrugBank XML (1.76 GB)

```
graph TD; A[DrugBank XML (1.76 GB)] --> B[XML Parsing (structured extraction)]; B --> C[Structured Parquet Tables<br/>drugs | interactions | targets | enzymes]; C --> D[Organized Data Storage (ready for feature engineering)];
```

**XML Parsing (structured
extraction)**

Structured Parquet Tables

drugs | interactions | targets | enzymes

**Organized Data Storage (ready for
feature engineering)**

DrugBank Dataset : Data Preprocessing and Feature Engineering

Data cleaning :

- Initial extraction produced ~1M rows (duplicated drug entries caused by XML extraction.
- later reduced to ~19K unique drugs. (one row per DrugBank ID)
- Standardized data types and handled missing values

Interaction Features

- Number of interactions as subject
- Number of interactions as affected drug
- Total interaction degree

Metabolic (CYP) Features

- number of enzyme records
- number of CYP-associated records
- unique CYP enzymes per drug

Safety Complexity Score

- combines interaction and metabolic complexity into one interpretable feature.

“Top Safety Complexity (not intrinsic danger)”

***	drugbank_id	name	total_interactions	n_unique_cyp	safety_complexity_score
8172	DB09061	Cannabidiol	3636.0	14.0	15.198914
1009	DB01026	Ketoconazole	3034.0	14.0	15.017967
241	DB00252	Phenytoin	4073.0	13.0	14.812381
12064	DB14009	Medical Cannabis	2356.0	14.0	14.765145
1154	DB01174	Phenobarbital	3884.0	13.0	14.764878
659	DB00675	Tamoxifen	3004.0	13.0	14.508033
12773	DB14737	Cannabinol	2422.0	13.0	14.292762
1214	DB01234	Dexamethasone	3510.0	12.0	14.163656
457	DB00470	Dronabinol	2896.0	12.0	13.971431
12685	DB14649	Dexamethasone acetate	2162.0	12.0	13.679251

The highest scoring drugs correspond to well-known interaction-heavy medications, suggesting the engineered safety complexity score aligns with real pharmacological knowledge.

DrugBank Dataset : Data Preprocessing and Feature Engineering

Final merged feature table → 19,830 drugs

Active subset (interaction OR CYP info) → 4,657 drugs

Train / Validation / Test split (70 / 15 / 15)

```
# summary
features[[
  "total_interactions",
  "n_unique_cyp",
  "safety_complexity_score"
]].describe()
```

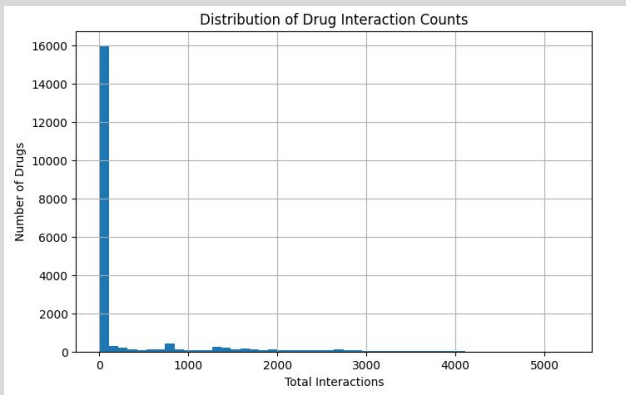
	total_interactions	n_unique_cyp	safety_complexity_score
count	19830.000000	19830.000000	19830.000000
mean	293.468381	0.214574	1.598675
std	739.955945	1.004001	3.080631
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000
max	5272.000000	14.000000	15.198914

```
# summary for features active
features_active[[
  "total_interactions",
  "n_unique_cyp",
  "safety_complexity_score"
]].describe()
```

	total_interactions	n_unique_cyp	safety_complexity_score
count	4657.000000	4657.000000	4657.000000
mean	1249.619498	0.913678	6.807328
std	1066.190746	1.911559	2.225461
min	0.000000	0.000000	0.500000
25%	252.000000	0.000000	5.581404
50%	1066.000000	0.000000	7.174724
75%	1920.000000	1.000000	7.930206
max	5272.000000	14.000000	15.198914

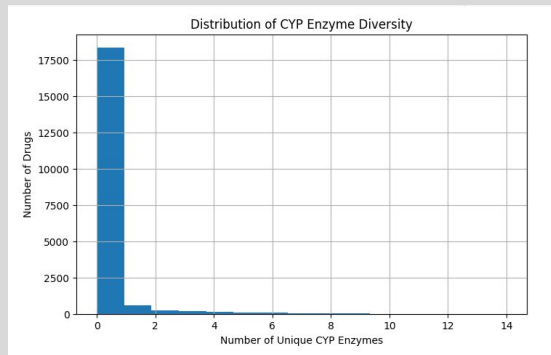
FULL dataset vs ACTIVE dataset

DrugBank Dataset: Exploratory Data Analysis (EDA)



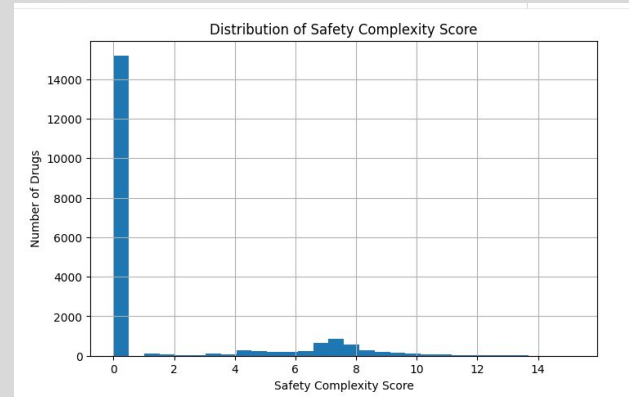
Plot 1- Interaction Distributions
Most drugs have few interactions.

Small number of drugs are highly connected.



Plot 2 CYP Enzyme Diversity
Many drugs involve few or no CYP pathways.

A smaller subset shows high metabolic complexity.



Plot 3- Safety Complexity Score
Combined feature separates high-complexity drugs from the majority.

Drugbank Dataset: Sparsity and Long-Tail Behavior

Sparsity = lots of missing or zero values in a dataset.

Long-tail distribution = most items have small values and a few items have extremely large values.

Why sparsity happens in pharmacological databases

Drug databases like DrugBank include:

- experimental compounds
- rare or discontinued drugs
- niche therapeutics
- newly studied drugs

Many of these:

- haven't been extensively studied,
- or have limited reported interactions.

Why long-tail behavior happens in pharmacology

Because science and medicine focus heavily on:

- common drugs
- older drugs
- clinically important drugs

These accumulate:

- more research
- more reported interactions
- more annotations

Sparsity refers to the large number of drugs with little or no recorded interaction or metabolism data, while long-tail behavior describes the small subset of highly studied drugs that account for most documented interaction

DrugBank Sources - <https://dev.drugbank.com/guides/drugbank/citing>

Knox C, Wilson M, Klinger CM, et al. *DrugBank 6.0: the DrugBank Knowledgebase for 2024*. Nucleic Acids Res. 2024 Jan 5;52(D1):D1265-D1275. doi: 10.1093/nar/gkad976.

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